

RESEARCH ARTICLE

The effects of carbonaceous inclusions and their distributions on dynamic failure processes in boron carbide ceramics

Arezoo Zare^{1,2}  | Pinkesh Malhotra^{1,2} | Qirong Yang^{3,4} | Kent Christian³  | Jun Du³ | Richard A. Haber³ | Matthew Shaeffer^{1,2} | K.T. Ramesh^{1,2}

¹Department of Mechanical Engineering, Johns Hopkins University, Baltimore, Maryland, USA

²Hopkins Extreme Materials Institute, Johns Hopkins University, Baltimore, Maryland, USA

³Department of Materials Science and Engineering, Rutgers University, Piscataway, New Jersey, USA

⁴Materials Science Division, Lawrence Livermore National Laboratory, Livermore, California, USA

Correspondence

Arezoo Zare, Department of Mechanical Engineering, Johns Hopkins University, Baltimore, MD 21218, USA.

Email: azare1@jhu.edu

Funding information

Army Research Laboratory, Grant/Award Number: W911NF-12-2-0022; U.S.

Department of Energy, Grant/Award Number: DEAC52-07NA27344

Abstract

Commercially available boron carbide ceramics typically have heterogeneous microstructures that contain distributions of processing-induced inclusions. The inclusions that are rich in carbon (i.e., carbonaceous) govern the underlying mechanisms of brittle fracture through wing crack formation, and thus dictate the mechanical response of the ceramic. In this study, we investigate the dynamic failure of five boron carbide ceramic materials with different inclusion populations. All of the materials were prepared by hot-pressing; four of these boron carbides contained different sizes and concentrations of carbonaceous inclusions, while one contained no carbonaceous inclusions. The heterogeneity distributions were characterized in some detail for statistical analysis using scanning electron microscopy and quantitative image analysis. A modified compression Kolsky bar setup with in situ ultra-high-speed microscopic imaging (10 million frames per second) was then used to study the influence of the inclusion distributions on the dynamic failure processes in these materials, at nominal high strain rates of 10^2 – 10^3 s⁻¹. The in situ ultra-high-speed microscopy highlighted the link between micro and macroscale failure processes and demonstrated that the carbonaceous inclusions are indeed the preferential sites for nucleation of wing cracks, as previously hypothesized based on post-mortem observations. The relative orientation of an inclusion with respect to the compression axis was shown to affect the likelihood that it would participate in crack nucleation. All of the ceramics were also found to have orientation-dependent peak compressive stress, regardless of the presence of carbonaceous inclusions, suggesting that grain orientation distributions are also important.

KEYWORDS

dynamic compression, high-throughput nanoindentation, Kolsky bar, microstructural design, processing-induced flaws, structural ceramics, wing crack

1 | INTRODUCTION AND BACKGROUND

Advanced structural ceramics (carbides, borides, nitrides, and oxides) offer many desirable properties including high hardness and elastic stiffness, high strength retention at elevated temperatures, low density, and good wear and corrosion resistance. As a result of the rapid development of improved processing technologies, such ceramics have found applications in the aerospace, defense, automotive, machining, nuclear, semiconductor, and energy storage industries.^{1–3} The ceramic components used in many of these applications (e.g., impact-resistant protective plates, high-speed cutting and grinding tools) are subjected to dynamic loading conditions^{3–5} which could result in brittle fracture and premature component failure. A clear understanding of the underlying mechanisms of brittle fracture and their correlation with the materials microstructure could provide a pathway toward designing structural ceramics that offer improved mechanical performance under dynamic loading conditions.

In this study, boron carbide (B_4C) is used as a model ceramic material to address the correlations between microstructure and dynamic brittle failure. Previous studies identified three key mechanisms that are activated within boron carbide under dynamic loading conditions: amorphization,^{6–9} massive fracture and fragmentation,^{10–14} and granular flow.¹⁵ Amorphization is the dominant mode of failure when high strain rate deformation and concurrent high confining pressure are present.¹⁶ Amorphization has been attributed to various underlying mechanisms including shear band formation,⁶ collapse of crystal structure,^{17,18} localized melting,¹⁹ dislocation/twin-mediated quasi-plasticity,^{20,21} and sliding of soft or amorphous interface phases at the grain boundaries of nanocrystalline ceramics (with typical average grain sizes of ≤ 100 nm).^{7,9} However, previous investigations of commercially available hot-pressed boron carbide ceramics (e.g., pressure-aided densified or PAD B_4C from CoorsTek, Inc. with an average grain size of $5\text{ }\mu\text{m}$) subjected to unconfined uniaxial or confined biaxial compression at nominal strain rates of 10^2 – 10^3 s^{-1} showed no evidence of amorphization under such stress states.^{11,13} Therefore, failure through amorphization is not the focus of this study. Granular flow is another failure mechanism that is mainly considered within the comminuted zone of a fragmented ceramic and is also not the focus of the present work. Under unconfined dynamic loads, massive fracture and fragmentation with little or no plastic deformation is the primary mode of failure in nearly all ceramics including boron carbide. Processing-induced microstructural defects or intrinsic flaws such as inclusions and pore clusters are the primary sites that result in crack initiation.

In boron carbide, dynamic failure is mediated by formation of cracks from microstructural inclusions that are rich in carbon (henceforth referred to as carbonaceous inclusions). Post-mortem examinations of fracture surfaces resulted from dynamic compression of PAD B_4C ceramics suggested that large carbonaceous inclusions undergo sliding and form tension cracks that deviate at sharp angles from the sliding plane. These tension cracks (commonly referred to as wing cracks) grow toward the direction of maximum compression and maximize the mode I stress intensity factor (K_I). In situ high-speed imaging of the failure process showed that structural failure occurs when a series of wing cracks extend and coalesce to split the material. Microstructural characterizations of PAD B_4C also showed that the carbonaceous inclusions are preferentially oriented with respect to the hot-pressing direction. This preferential orientation was postulated to introduce anisotropy in the dynamic compressive strength of boron carbide.^{12,14} Even though the contribution of carbonaceous inclusions to wing crack formation is established, the role of inclusion distributions on micro-crack nucleation and macro-crack development has remained unexplored. During dynamic loading, the rate of increase of the load can outpace the finite growth velocity of the wing cracks initiated from the large carbonaceous inclusions.²² Therefore, the carbonaceous inclusions next in size can also be activated and contribute to failure. Consequently, the entire distribution of carbonaceous inclusions is an important consideration when wing cracks dominate the failure.^{22–25}

Building on these previous works, and taking advantage of recent developments that allow for statistical and more detailed analysis of microstructure prior to and during deformation, this study presents a systematic investigation of the relationships between the distributions of carbonaceous inclusions and dynamic compressive failure through the wing crack mechanism. Our findings can help guide the design of new boron carbide ceramics with improved dynamic mechanical performance, and also help to understand the dynamic failure of other types of structural ceramics (e.g., SiC , Al_2O_3) and brittle solids (e.g., geomaterials) that fail through formation of wing cracks from intrinsic flaws.

2 | PREPARATION OF MATERIALS AND SPECIMENS

Boron carbide materials were prepared by hot-pressing a mixture of commercially available powders of boron carbide (H.C. Starck, Grade HD20, with an average particle size of $0.5\text{ }\mu\text{m}$ and a purity of 97.6%), graphite (Morgan, Grade G-23 with a D_{50} particle size of $6\text{--}8\text{ }\mu\text{m}$ henceforth referred to as small or Grade G-46 with a D_{50} particle size

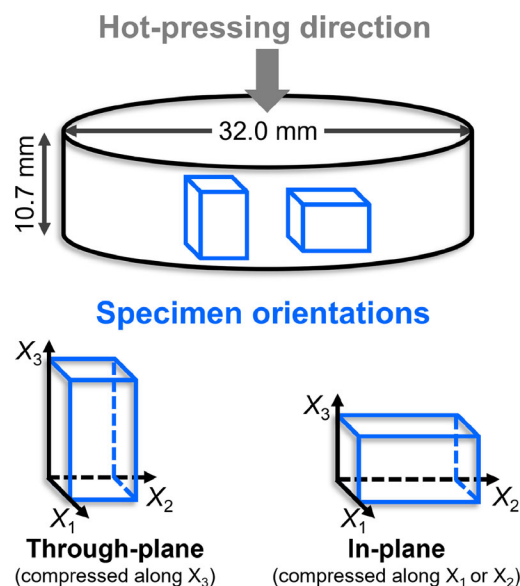
TABLE 1 Details of powder mixture compositions and hot-pressing conditions used to prepare boron carbide materials.

Targeted microstructure	Initial powder mixture	Hot-pressing conditions
S-inclusion L-spacing (SILS)	B_4C + 1 wt.% small graphite (6–8 μm D50)	2100°C, 35 MPa
S-inclusion S-spacing (SISS)	B_4C + 5 wt.% small graphite (6–8 μm D50)	2100°C, 35 MPa
L-inclusion L-spacing (LILS)	B_4C + 1 wt.% large graphite (40–75 μm D50)	2100°C, 35 MPa
L-inclusion S-spacing (LISS)	B_4C + 5 wt.% large graphite (40–75 μm D50)	2100°C, 35 MPa
No free carbon (NFC)	B_4C + 2 wt.% amorphous B	2150°C, 24 MPa

of 40–75 μm henceforth referred to as large), and amorphous boron (H.C. Starck, Grade I, with an average particle size of 1 μm and a purity of 95%).

The objective was to obtain materials with different inclusion sizes as well as different inclusion spacings. To control the size of the carbonaceous inclusions, the boron carbide powder was mixed with either the small (S) or large (L) graphite powders (shown in Figure S1). The spacing of the carbonaceous inclusions was controlled by varying the concentration of the graphite in the powder mixtures from 1 or 5 wt.%. Increasing the concentration of the graphite powder was expected to result in a smaller spacing between the carbonaceous inclusions after hot-pressing. The graphite powder was mixed with boron carbide powder in a Nalgene bottle with SiC media and ethanol for 30 min. The powder mixture was dried in a vacuum oven over night and gently deagglomerated using mortar and pestle before sintering. Four boron carbide disks (32.0 mm in diameter and 10.7 mm in height) were hot-pressed under the same conditions detailed in Table 1. These materials will henceforth be denoted by the *targeted* sizes of the resulting carbonaceous inclusions and the spacings between them (e.g., S-inclusion with L-spacing or SILS; note that the *targeted* sizes and spacings are NOT the actual sizes and spacing achieved).

To prepare a baseline material that contains no carbonaceous inclusions, one additional hot-pressed tile (4.2 inch $\times$ 4.2 inch $\times$ 14.8 mm) was also prepared from a mixture of boron carbide and amorphous boron powders. The additional amorphous boron is used to react with the residual free (unbounded) carbon contained in the commercial boron carbide powder, which was determined to be ~ 0.55 wt.%.²⁶ This material will henceforth be denoted as no free carbon (NFC). The as-received commercial boron carbide and amorphous boron powders were soaked in methanol for 24 h to dissolve excess surface boron oxide and then decanted. The washed powder was then vacuum dried to avoid further oxidation. The powder mixtures were rotationally mixed with SiC media in ethanol for 1 h, vacuum dried at 70°C for 12 h, and then gently crushed to deagglomerate before hot-pressing under the conditions specified in Table 1. The density of all the hot-pressed mate-

**FIGURE 1** Schematic representation of a hot-pressed material and orientations of machined prismatic specimens and their relations to the hot-pressing direction. Note that the X_1 and X_2 axes are considered to be equivalent.

rials was evaluated using Archimedes' principle in water and found to be $\geq 99.2\%$ the theoretical density of B_4C .

The hot-pressed materials were machined to prismatic specimens for mechanical characterization, with rectangular cross-sections and nominal dimensions of $3.5 \times 4.0 \times 5.3$ mm³, and using a 1200 grit final finish for the surface preparation. The choice of the specimen geometry was guided by the need to obtain a reasonable depth of field for in situ visualization of dynamic failure processes during mechanical testing (further discussed in Section 3.5). The specimens were cut in accordance with the coordinate system $X_1 \times X_2 \times X_3$ shown schematically in Figure 1, where X_3 is parallel to the hot-pressing direction and X_1 and X_2 are the principal directions lying in the hot-pressed surface. Two sets of specimens were prepared from each material: with their 5.3 mm edges being either parallel ("through-plane") or perpendicular ("in-plane") to the X_3 -axis.

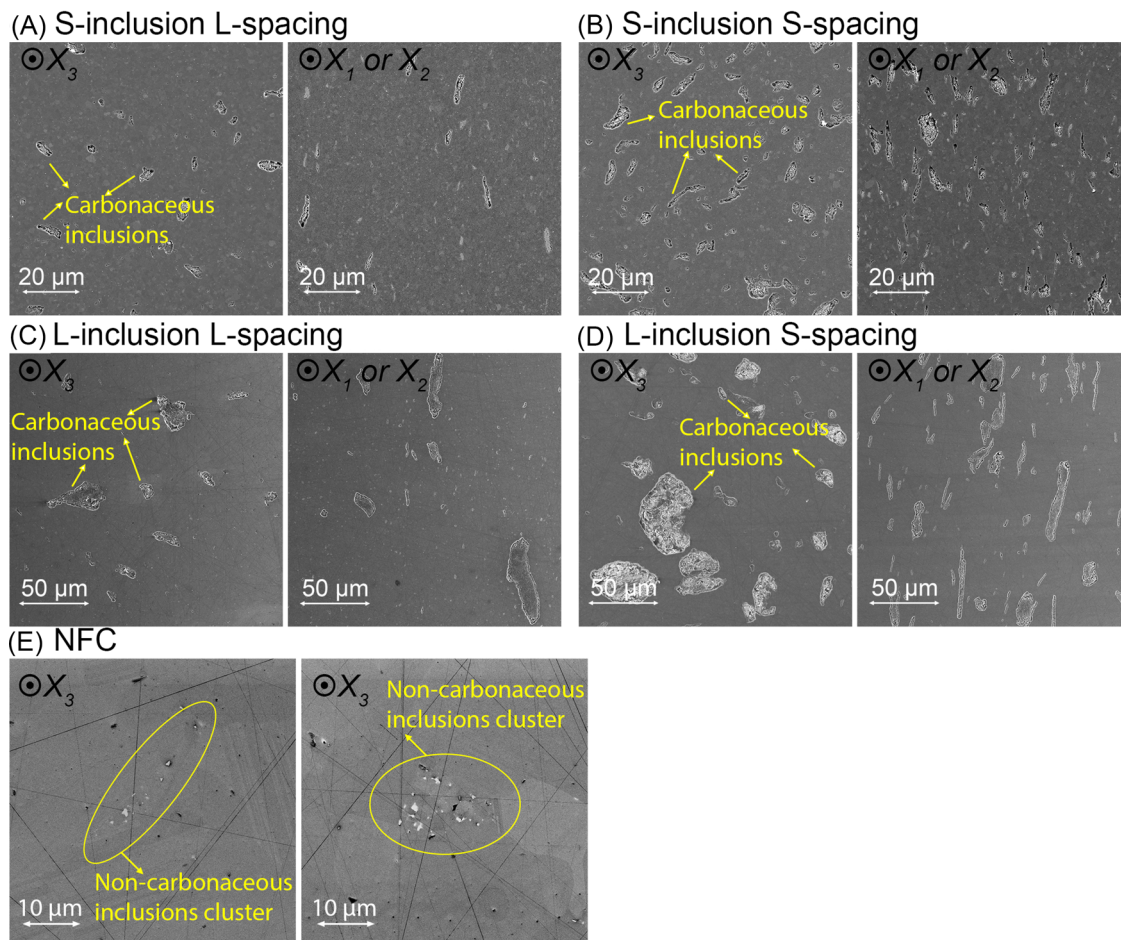


FIGURE 2 SEM micrographs of hot-pressed B_4C materials in the two studied orientations. SEM, scanning electron microscopy.

3 | EXPERIMENTAL METHODS

3.1 | Microstructural characterization

For each tested material, two faces of the prismatic specimens (one with a surface normal of X_3 and another one with a surface normal of X_1 or X_2) were mechanically polished to a mirror-finish end using diamond lapping films, and their microstructures were characterized using scanning electron microscopy (SEM). Figure 2 shows representative SEM micrographs for each material in the two studied orientations. As seen in Figure 2A–D, the four S- and L-inclusion *targeted* microstructures contain different sizes and concentrations of microstructural inclusions. The inclusions also appear to have more elongated cross-sections when viewed normal to X_1 or X_2 . Energy dispersive spectroscopy (EDS) characterization of one of the detected inclusions in the L-inclusion S-spacing (LISS) *targeted* microstructure (shown in Figure S2A) confirmed that the observed inclusions are rich in carbon with trace amounts of oxygen present at the inclusion periphery. These inclusions will henceforth be referred to as carbonaceous inclusions and are considered to be the primary type

of inclusions present in the SILS, SISS, LILS, and LISS materials. SEM micrographs of the NFC material, seen in Figure 2E, show sporadic clusters of sub-micron inclusions embedded in the matrix. EDS characterization of the inclusion clusters (shown in Figure S2B) showed that they do not contain carbon and consists of oxygen, silicon, aluminum, and titanium. These non-carbonaceous inclusion clusters are the only type of inclusions observed in the NFC material. We note that microstructural characterization of the NFC material was only attempted on the surfaces normal to X_3 .

Grain sizes of the B_4C materials were evaluated using the line intercept method from select SEM micrographs collected from surfaces normal to X_3 . All these B_4C materials were found to have comparable grain sizes of 0.8–1.0 μm .

3.2 | Statistical analysis of carbonaceous inclusions

A MATLAB script was written to segment the collected SEM micrographs in order to statistically analyze four

attributes of the carbonaceous inclusions: size, spacing, shape, and orientation. First, adaptive thresholding was used to convert the grayscale images to monochrome followed by an area opening operation to remove connected objects with pixels fewer than a minimum value to reduce noise. To connect the individual objects identified by thresholding, a flat disk-shaped structuring element was used to perform a dilation and erosion sequence. Any remaining holes were then filled and the exterior boundaries of the identified objects were traced. The minimum number of connected pixels used in the area opening operation and the radius of the structuring element were adjusted for each micrograph accordingly with the typical values ranging from 20–35 and 1–5 respectively for S-inclusion materials and 20–25 and 3–8 respectively for L-inclusion materials. Examples of the obtained binary images versus the original grayscale micrographs for the S- and L-inclusion materials are shown in Figure S3. All the SEM micrographs were collected at a magnification of 2768× and 1384× for materials with S- and L-inclusions, respectively. The selected magnifications allowed for capturing high-resolution information while maintaining a reasonably large field of view.

For each material and orientation, 13–28 distinct SEM micrographs were segmented and the obtained binary images were used to compute the major and minor axes of an ellipse with the same normalized second central moment as each identified object. The major axis length and the ratio of major to minor axes were used to represent the inclusion size and shape, respectively. The spacing between neighboring inclusions was computed by finding the distance between the center of mass (centroid) of each identified object with its closest neighbor. Only one closest neighbor was considered in our spacing analysis. Inclusion orientation was evaluated by finding the angle between the normal to an inclusion's major axis and a reference in-plane orientation in the field of view. For each SEM micrograph, the reference in-plane orientation was selected to be parallel to the compression axis later used to dynamically test the specimens. The areal number density of the inclusions was also computed based on the number of detected carbonaceous inclusions and the total studied area.

3.3 | Characterization of quasi-static mechanical response

An iNano[®] nanoindenter with an InForce50 actuator (Nanomechanics Inc., now KLA Corporation) was used to obtain spatial maps of indentation modulus and hardness of the four S- and L-inclusion *targeted* microstructures. A high-speed nanoindentation technique (Nanoblitz3D[®]) was used to perform an array of 40 × 40 indents on each

specimen. Based on previous measurements on B₄C single crystals,²⁷ the spacing between neighboring indents was set to 10× the indentation depth (~3 μm). All indentations were performed using a diamond Berkovich indenter tip under a maximum applied load (F_{max}) of 50 mN resulting in a maximum indentation depth (h_{max}) of ~300 nm. The initial stage of the unloading curve from each indentation was fit to a power law which then allowed for the determination of contact stiffness (S) using

$$S = \frac{dF}{dh} \quad (1)$$

where F and h are the indentation load and depth, respectively. The indentation modulus (E_{IT}) and hardness (H_{IT}) were subsequently obtained using the following equations:

$$E_{IT} = \frac{1 - \nu_s^2}{\frac{1}{E_r} - \frac{1 - \nu_i^2}{E_i}} \quad (2)$$

$$H_{IT} = \frac{F_{max}}{A_p} \quad (3)$$

where ν_s is the Poisson's ratio of the specimen (assumed to be 0.17²⁸), ν_i and E_i are the Poisson's ratio and Young's modulus of the indenter (0.07 and 1141 GPa, respectively), A_p is the projected contact area, and E_r is the reduced elastic modulus of the specimen obtained using

$$E_r = \frac{\sqrt{\pi}}{2} \frac{S}{\sqrt{A_p}} \quad (4)$$

The projected contact area was determined through knowledge of the indenter's area function and the depth over which the indenter and specimen were in contact. The area function of the indenter was calibrated by performing indentations on a fused silica reference specimen. The k -means data clustering algorithm was employed to partition the indentation map data into separate populations in order to determine the modulus and hardness of the B₄C matrix and carbonaceous inclusions.

3.4 | Characterization of dynamic mechanical response

Dynamic response was studied under uniaxial loading conditions at nominal strain rates of 10²–10³ s⁻¹ using a modified compression Kolsky bar setup similar to that described in ref. 10. The bars and projectile were made of C350 maraging steel with a density of 8.1 g/cm³ and a Young's modulus of 200 GPa and had a diameter of 12.7 mm. A pulse-shaping technique was used to impose a triangular loading pulse on the specimen using a combined copper and graphite pulse shaper between the

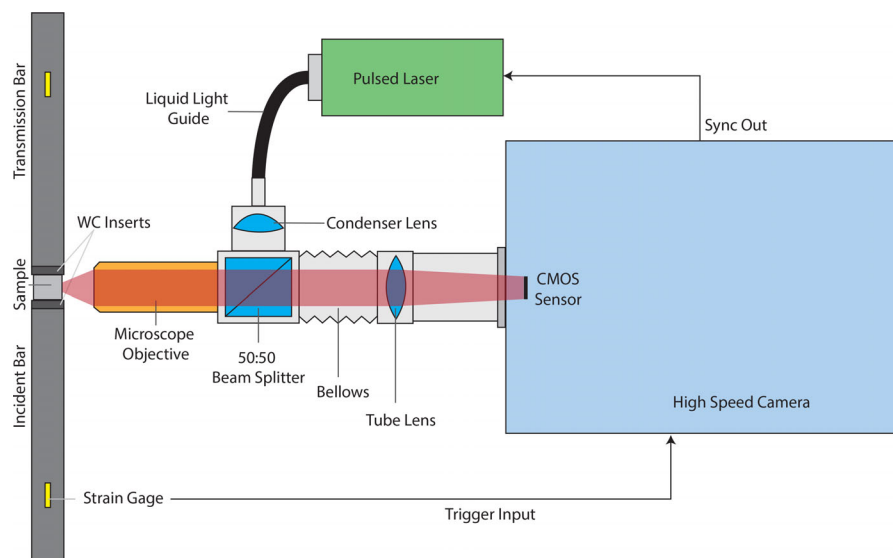


FIGURE 3 Schematic of ultra-high-speed microscopy set-up integrated with a Kolsky bar (not drawn to scale). The ultra-high-speed camera is triggered using the strain gage signal from the incident bar. The pulsed laser is synchronized with the high-speed camera exposure using sync out signals from the camera. The reflected light path is shown in red.

projectile and the incident bar. This ensured the loading rate remained relatively constant and that stress equilibration had been achieved in the specimen prior to failure. After the pulse was shaped, it propagated down the incident and transmission bars and its magnitude was recorded by strain gages. Impedance-matched sacrificial inserts (i.e., platens) made from a tungsten carbide core confined by a heat-shrunk Ti-6Al-4V collar were placed between the bar ends and the specimen to prevent damage in the steel bars. The interfaces between the inserts and specimen were lubricated with molybdenum grease to minimize interfacial friction. Between 3–10 specimens were tested in the Kolsky bar for each material and orientation, but some of the tests were deemed invalid due to strain gage failure.

3.5 | In situ observations of dynamic failure processes

The prismatic geometry of the specimens allowed for in situ visualization of the failure process during the Kolsky bar experiments. A Kirana Ultra High-Speed Camera (Tring, Hertfordshire, UK) operating at 5 million frames per second was used for visualization and the specimen surface was illuminated with a pulsed laser. For select experiments, high-speed microscopic imaging was combined with the Kolsky bar (see Figure 3) to carry out an in situ study of microscopic deformation mechanisms while simultaneously recording the macroscopic material behavior. A microscope similar to the one employed by Malhotra et al.²⁹ was integrated in front of a Shimadzu high-speed

camera to provide a spatial resolution of $3\ \mu\text{m}$ per pixel at a frame rate of 10 million frames per second. The optical train consists of an infinity-corrected $10\times$ long working distance objective with a numerical aperture (NA) of 0.28, followed by a 50:50 beam splitter and a lens tube. A high NA objective allows to keep the diffraction limit to a small value ($\frac{\lambda}{2\text{NA}} = 1.14\ \mu\text{m}$ for $\lambda = 640\ \text{nm}$) and hence achieve a high spatial resolution, which in this case is limited by the finite pixel size of the camera. A 400 W pulsed laser, placed at a right angle to the optic axis of the microscope, was used for in-line illumination. Light from the laser is directed to the sample by the beam splitter via the microscope objective. Since all the light from the pulsed laser is directed on to the sample, such an arrangement makes more efficient use of illumination as compared to the use of multiple halogen lamps. Pulses as short as 10 ns from the pulsed laser help reduce motion blur, which is not possible through the use of continuous illumination. Output pulses from the camera were used to synchronize the illumination pulses with the acquired images. The output pulses from the camera were also used to synchronize the camera images with the stress–time history computed from the signals recorded by the strain gages on the bars.

4 | RESULTS

We begin by presenting a detailed quantitative description of the distributions of the carbonaceous inclusions within these materials, with a view to providing a dataset that can be used for developing and assessing both microstructural and mechanical models.

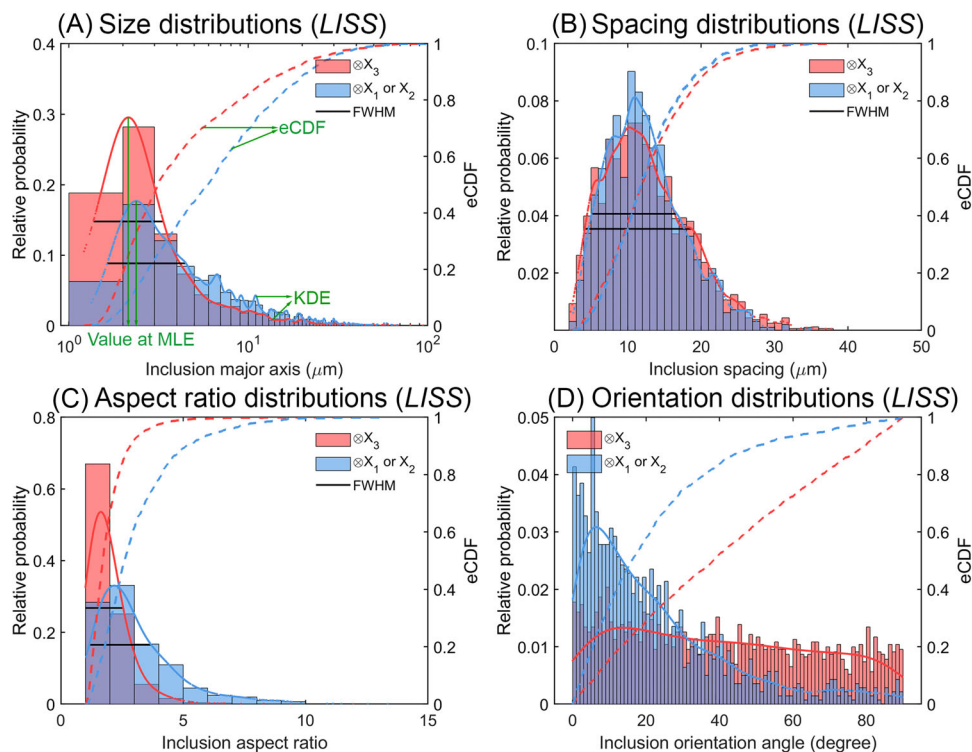


FIGURE 4 Distributions of the (A) size, (B) spacing, (C) aspect ratio, and (D) orientation of the carbonaceous inclusions in the large-inclusion small-spacing *targeted* microstructure. For each distribution, the solid and dashed curves show the kernel density estimate (KDE) of the probability density function and the empirical cumulative distribution (eCDF) function.

4.1 | Distributions of carbonaceous inclusion attributes

Figure 4 shows the representative distributions of the size, spacing, aspect ratio, and orientation of the carbonaceous inclusions, obtained for the LISS *targeted* microstructure. The distributions of inclusion attributes in the SILS, SISS, and LILS materials are also shown in Figures S4–S6. For a given attribute, the distributions are shown in either the through-plane (in blue) or the in-plane (in red) cross-sections. The histograms are plotted against relative probability, which is defined as

$$v_i = \frac{c_i}{n} \quad (5)$$

where v_i is the relative probability, c_i is the number of elements in the i th bin, and n is the total number of elements in the input data. For each material orientation (through-plane or in-plane), we also plot the corresponding empirical cumulative distribution functions (eCDF) as dashed red or blue lines, with the eCDF obtained from

$$\text{eCDF} = \sum_{j=1}^i \frac{c_j}{n}. \quad (6)$$

For each distribution, a normal kernel smoothing function ($K(\cdot)$) with a bandwidth of h was used to obtain an estimate $\hat{f}_h(x)$ of the probability density function ($f(x)$) using

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_j}{h}\right) \quad (7)$$

with $x_1, x_2, \dots, x_n$ representing individual data points. Each smoothing function was defined as

$$K(\cdot) = \frac{1}{h\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-x_j}{h}\right)^2}. \quad (8)$$

The obtained kernel density estimate (KDE) probability curves are superimposed on the histograms.

For all the materials and orientations, the size and spacing distributions of carbonaceous inclusions are found to be right skewed. To rule out bias toward larger inclusions in our analysis, the maximum achievable spatial resolution of the SEM micrographs was estimated using the Nyquist limit.³⁰ The pixel-limited spatial resolutions for the S- and L-inclusion materials (0.2 and 0.4 μm , respectively) were found to be smaller than the lower range of computed values. The value corresponding to the maximum likelihood estimate (MLE) of the KDE probability function and the range of values that correspond to full width at half

TABLE 2 Summary of the maximum likelihood estimate (MLE) and full width at half maximum (FWHM) values of the size of carbonaceous inclusions in different materials.

Targeted microstructure	Orientation	Size MLE	Size FWHM	Fitted PDF (exp,shp,loc,scl*)
S-inclusion L-spacing	$\odot X_3$	1.5 μm	1.2 μm	Weibull (1.00,−1.85,0.58,0.97)
S-inclusion L-spacing	$\odot X_1 \text{ or } X_2$	1.7 μm	1.6 μm	Weibull (49.31,0.32,0.83,0.01)
S-inclusion S-spacing	$\odot X_3$	1.5 μm	1.3 μm	Weibull (16.41,0.32,0.87,0.04)
S-inclusion S-spacing	$\odot X_1 \text{ or } X_2$	1.5 μm	1.5 μm	Weibull (11.84,0.37,0.83,0.09)
L-inclusion L-spacing	$\odot X_3$	2.2 μm	2.0 μm	Weibull (1.00,−1.45,0.92,1.52)
L-inclusion L-spacing	$\odot X_1 \text{ or } X_2$	2.1 μm	2.3 μm	Weibull (1.00,−1.41,0.90,1.27)
L-inclusion S-spacing	$\odot X_3$	2.1 μm	2.3 μm	Weibull (1.00,−1.06,1.05,1.55)
L-inclusion S-spacing	$\odot X_1 \text{ or } X_2$	2.4 μm	2.9 μm	Weibull (4.54,0.46,1.32,1.04)

The fitted probability distribution functions and their parameters are also shown. *Exponentiation parameter, shape parameter, location, scale.

TABLE 3 Summary of the maximum likelihood estimate (MLE) and full width at half maximum (FWHM) values of the spacing of carbonaceous inclusions in different materials.

Targeted microstructure	Orientation	Spacing MLE	Spacing FWHM	Fitted PDF (exp,shp,loc,scl*)	No. density
S-inclusion L-spacing	$\odot X_3$	3.3 μm	3.8 μm	Weibull (1.94,1.30,1.34,2.34)	2.2E+10/m ²
S-inclusion L-spacing	$\odot X_1 \text{ or } X_2$	4.0 μm	4.9 μm	Weibull (1.80,1.30,1.32,3.38)	1.6E+10/m ²
S-inclusion S-spacing	$\odot X_3$	4.4 μm	4.2 μm	Chi-squared (NA,12.13,0.40,0.39)	1.7E+10/m ²
S-inclusion S-spacing	$\odot X_1 \text{ or } X_2$	3.7 μm	3.6 μm	Weibull (1.48,1.72,1.27,2.90)	2.4E+10/m ²
L-inclusion L-spacing	$\odot X_3$	10.6 μm	16.3 μm	Weibull (1.12,1.67,2.82,11.96)	2.0E+9/m ²
L-inclusion L-spacing	$\odot X_1 \text{ or } X_2$	8.6 μm	10.9 μm	Weibull (1.19,1.55,1.97,8.28)	3.4E+9/m ²
L-inclusion S-spacing	$\odot X_3$	10.2 μm	14.7 μm	Weibull (1.03,1.79,2.13,11.70)	2.7E+9/m ²
L-inclusion S-spacing	$\odot X_1 \text{ or } X_2$	10.0 μm	11.6 μm	Log-normal (NA,0.30,−5.06,16.65)	2.7E+9/m ²

The fitted probability distribution functions and their parameters are also shown. *Exponentiation parameter, shape parameter, location, scale.

of the maximum probability (FWHM) were adopted as the parameters to compare the center and spread of the distributions in different microstructures. The obtained MLE and FWHM values for the size and spacing of carbonaceous inclusions are summarized in Tables 2 and 3, respectively.

The MLE values of inclusion sizes reported in Table 2 are considerably smaller than the reported *D*50 values of the graphite powders used in hot-pressing (6–8 μm for the SILS and SISS *targeted* microstructures and 40–75 μm for the LILS and LISS *targeted* microstructures, from Table 1). It is likely that the loosely-bonded graphite platelets were reduced in size by the harder SiC media and boron carbide powder during the mixing process, and that the larger platelets were more prone to being crushed by the unintended milling process. As a result, a significant overlap exists between the inclusion sizes of the SILS and SISS materials with those of the LILS and LISS. The measured distributions of inclusion sizes were also fitted to various common probability distribution functions (including Weibull, normal, log-normal, gamma, binomial, uniform, Poisson, Rayleigh, Cauchy, powerlaw, and Pareto) and the Weibull distribution was found to be the best fit for all cases (using a cut-off *p*-value of 0.01). The following form was assumed for the Weibull distribution

$$g(x) = a|c|[1 - \exp(-x)^c]^{a-1} \exp(-x^c)x^{c-1},$$

$$\text{for } x > 0 \text{ and } a > 0 \quad (9)$$

with *a* and *c* representing the exponentiation and shape parameters, respectively. The characteristic parameters as well as the location and scale of the fitted PDFs are also summarized in Table 2.

As seen in Table 3, the MLE and FWHM of inclusion spacings for the SILS and SISS *targeted* microstructures are nearly half of those in the LILS and LISS materials. This is consistent with the measured areal number density of the inclusions in the former being an order of magnitude larger than the latter. However, the MLE and FWHM of the spacings in the SILS and SISS *targeted* microstructures are comparable and do not scale with the increase of graphite powder concentration in the initial powder mixtures from 1 to 5 wt.%. Similar observations are made when comparing the spacing values of the LILS and LISS materials. Further work is required to determine the cause of this discrepancy. Fitting the spacing distributions to common PDFs suggests that a Weibull distribution with the form described in Equation (9) best describes the data for most microstructures. Chi-squared and log-normal distributions were found to be a better fit than Weibull for two of

the microstructures. The Chi-squared distribution has the following form:

$$g(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} \exp\left(-\frac{x}{2}\right), \text{ for } x > 0 \text{ and } k > 0, \quad (10)$$

where $\Gamma(k/2)$ denotes the gamma function of the integer k . The form of the log-normal distribution is

$$g(x) = \frac{1}{sx\sqrt{2\pi}} \exp\left(-\frac{\log^2(x)}{2s^2}\right), \text{ for } x > 0 \text{ and } s > 0, \quad (11)$$

with s representing the shape parameter. The characteristic parameters as well as the location and scale of the fitted PDFs are summarized in Table 3.

The MLE values of aspect ratio were found to be comparable in nearly all the microstructures with the values in the range of 1.5–1.7 μm . For all materials, the aspect ratio distributions are further skewed to the right when inclusions are viewed on planes normal to X_1 or X_2 (as seen in Figure 4C) and the FWHM ranges were also found to be wider by 0.1–0.5. This implies that the inclusion cross-sections are more elongated on planes normal to X_1 or X_2 and more circular on planes normal to X_3 which agrees with previous reports on PAD B₄C that the carbonaceous inclusions are disk-shaped and tend to orient themselves perpendicular to the hot-pressing direction (i.e., X_3 -axis).¹⁰

Inclusion orientations were characterized by finding the angle between the normal to the major axis of each inclusion and the X_1 , X_2 , X_3 directions, and then transformed in terms of the angle between the normal and the compression loading axis used to evaluate the dynamic mechanical response of the materials. For all materials, the orientation distributions appear to be more or less uniform on planes normal to X_3 . In contrast, the distributions are right skewed on planes normal to X_1 or X_2 (as seen in Figure 4D with the MLE values of angle ranging between 6°–10°. For an intrinsic flaw in a brittle solid, mode I stress intensity factor is maximized when the flaw normal is favorably oriented with respect to the compression axis. The critical angle corresponding to this favorable orientation depends on the coefficient of friction between the sliding faces of the flaw.³¹ For boron carbide, a coefficient of friction of 0.6 is commonly assumed which corresponds to a critical angle of 60°. ^{22,32} Based on the KDE probability plots, the probability of finding inclusions that satisfy the 60° orientation angle is higher on planes normal to X_3 ; however, this distribution does not tell us if the flaws are big enough to trigger fracture.

To probe the correlations between the orientations of inclusions and their sizes, the distributions shown in Figure 4A,D were used to obtain the bivariate KDE of the probability density functions shown in Figure 5. The

orientation and size of the inclusions are found to be correlated with the probability of finding inclusions that satisfy the 60° orientation angle decreasing as the size of the inclusions increases. This observation will be subsequently linked to the failure processes observed during Kolsky bar experiments. A similar correlation was also observed between the orientation and size of inclusions in the S-inclusion S-spacing *targeted* microstructure (see Figure S7).

4.2 | Mechanical behavior under quasi-static loading

Figure 6A,C shows the spatial maps of indentation modulus and hardness obtained by performing an array of 40×40 indents on the material with LISS *targeted* microstructure. The carbonaceous inclusions can be readily distinguished from the boron carbide matrix due to their significantly lower mechanical properties. Similar maps were obtained from the other materials and used to compute the average values of indentation modulus and hardness shown in Table 4. Given that the indentation arrays were performed over regions that included both the boron carbide matrix and carbonaceous inclusions, the reported averages represent the properties of the composite material. This also explains the large standard deviations obtained. Even though reports on the Young's modulus of boron carbide using bulk techniques exist (e.g., Vargas-Gonzalez et al.³³ obtained a Young's modulus of 445.5 GPa for PAD B₄C using ultrasound), a comparison between the reported modulus and those obtained in this study was not attempted due to the inherent differences in boundary conditions between bulk measurement techniques and nanoindentation. It is also noted that indentation hardness is dependent on the indenter geometry and load and is not considered to be a material property.³⁴

Clustering the data from indentation maps into two populations was attempted to obtain the properties of a hard and a soft consistent phase. Only for the LISS material, clustering yielded two populations with statistically different values of modulus and hardness. This is likely due to insufficient number of data points from the carbonaceous inclusions on the other materials. Cluster maps of indentation modulus and hardness of the material with LISS *targeted* microstructure (shown in Figure 6B,D) predict the spatial position of the carbonaceous inclusions reasonably well. However, the modulus and hardness obtained for the soft constituent phase (summarized in Table 5) are significantly higher than the range of reported properties for graphite flakes (modulus of 10–50 GPa and hardness of 2–2.4 GPa^{35,36}). Boundary effects, either from the surrounding boron carbide matrix or the underlying boron

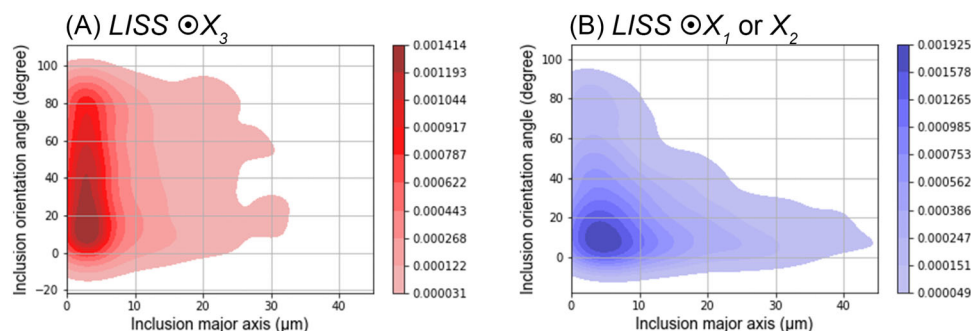


FIGURE 5 Bivariate KDE of the probability density functions for the orientation and size of the inclusions in the large-inclusion small-spacing *targeted* microstructure. KDE, kernel density estimate.

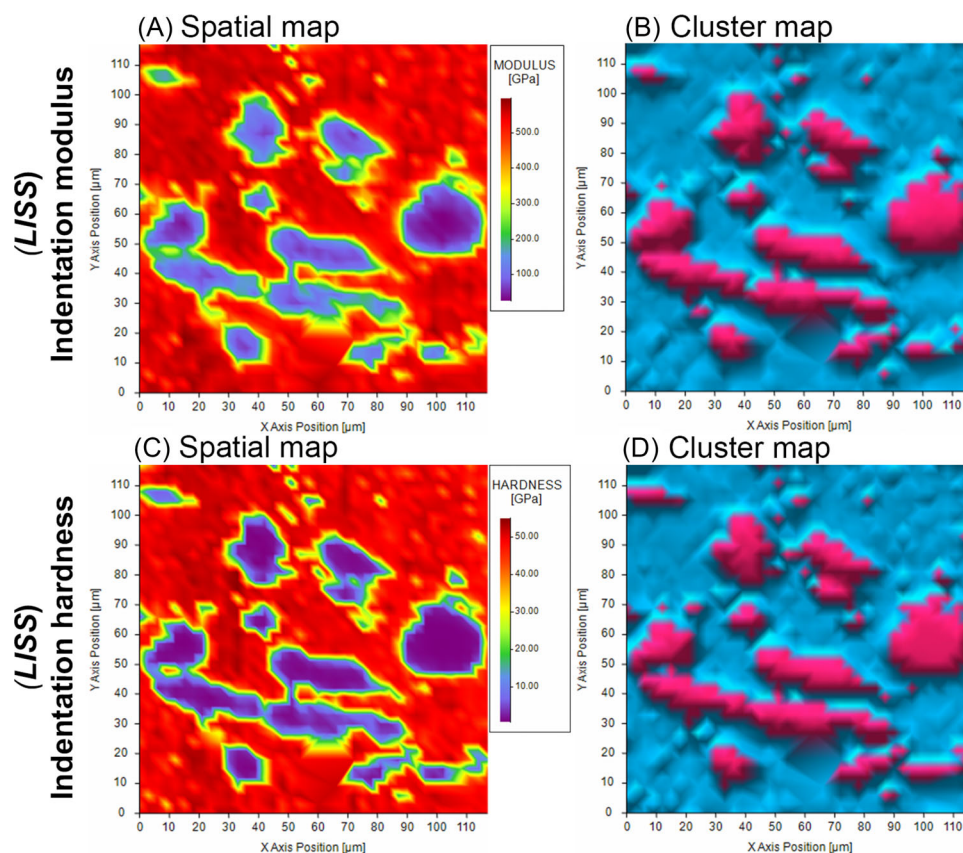


FIGURE 6 Spatial and cluster maps of indentation modulus and hardness for the L-inclusion S-spacing *targeted* microstructure.

TABLE 4 Summary of the composite modulus and hardness of the boron carbide materials obtained from nanoindentation mapping.

Targeted microstructure	Orientation	Modulus (GPa)	Hardness (GPa)
S-inclusion L-spacing	$\odot X_1 \text{ or } X_2$	523 ± 59	47 ± 8
S-inclusion S-spacing	$\odot X_1 \text{ or } X_2$	431 ± 87	42 ± 13
L-inclusion L-spacing	$\odot X_1 \text{ or } X_2$	468 ± 60	46 ± 8
L-inclusion S-spacing	$\odot X_1 \text{ or } X_2$	433 ± 176	37 ± 19

TABLE 5 Indentation modulus and hardness of the constituent phases of the boron carbide materials obtained from *k*-means clustering of nanoindentation maps of the L-inclusion S-spacing *targeted* microstructure.

Constituent phase	Fraction (%)	Modulus (GPa)	Hardness (GPa)
Hard constituent phase	74	530 ± 47	49 ± 4
Soft constituent phase	26	148 ± 78	7 ± 7

carbide substrate, are a plausible explanation for this discrepancy. Increasing the size of the indentation array could provide more accurate measurements of the properties of the carbonaceous inclusions by deconvolving their properties from the boundaries. We also note that the obtained properties of the hard constituent phase are in good agreement with the reported modulus and hardness for the boron carbide matrix of PAD B₄C obtained from nanoindentation at comparable indentation loads (550–557 GPa and 44–45 GPa, respectively¹³).

4.3 | Mechanical behavior under dynamic compression

4.3.1 | Dynamic failure processes

The high-speed microscopy setup allows us to image the dynamic failure process with unparalleled resolution. The results from dynamic uniaxial compression in the Kol-sky bar are typified by the evolution of the stress with time shown in Figure 7 together with the corresponding frames of the recorded high-speed video from an experiment on the LISS material. For this experiment, the compression axis was parallel to the X_3 -axis and imaging was performed on a plane normal to X_1 or X_2 . The imaged surface was mechanically polished to a mirror-finish to allow for visualization of carbonaceous inclusions and their role in the dynamic failure processes, using a 10× microscope objective and laser illumination. In the recorded frames, the features with a darker contrast are carbonaceous inclusions. Consistent with the distributions of inclusion orientation discussed in Section 4.1, the majority of the observable inclusions are oriented perpendicular to the compression axis. We also note that the areal field of view is ~5% of the total area of the surface and specimen edges/corners cannot be viewed. The peak (“maximum”) compressive stress was achieved at time t_5 and has a magnitude of 2.5 GPa. Various deformation processes are observed at times t_1 – t_4 even before the peak stress is achieved. First, several clouds of fine ejecta were observed, with the clouds appearing to become diffuse and larger in diameter at later times. The first instance of ejecta cloud formation was observed at time t_1 around the circled region. The ejecta can be better visualized in the video recording of this experiment which can be accessed by following the link provided for Video 1 in Section S1. At time t_2 , the first axial microcracks were seen to initiate from the carbonaceous inclusions. An example of such an initiation from one of these inclusions is highlighted in the insets shown on the right. Comparing the insets at times t_2 and t_1 shows microcracks that are formed on both sides of the inclusion. Tracking these microcracks at later times

suggests that their propagation had been arrested at other flaws, with the cracks still being visible and isolated at the time of the peak stress (see the inset at time t_5). At time t_3 , a set of axial and transverse microcracks are formed in one region resulting in the formation of a microscale fragment, and a subsequent localized loss of material as seen circled in time t_4 . At time t_5 corresponding to the peak stress, a large number of axial microcracks can be seen, each connected to one or more carbonaceous inclusions, together with a few regions where fragments have been ejected. As the stress collapses at time t_6 , rapid growth of damage has set in the form of axial and transverse cracks, and fragmentation continues to grow. Eventually, several parallel axial microcracks are formed from neighboring inclusions which rapidly coalesce into an unstable axial macrocrack spanning across the field of view. This span-wise crack results in the formation of larger columnar fragments.

The stress rate ($\dot{\sigma}$), determined as the slope of the stress–time plot between 10% and 90% of the peak stress (highlighted in green), was estimated to be 140 MPa/ μm in this experiment. The corresponding strain rate ($\dot{\epsilon}$) can be estimated by dividing the stress rate by the composite modulus of the material reported in Table 4. For the example shown in Figure 7, the strain rate was estimated as 347 s^{−1}.

To investigate the effects of orientation, the same LISS material was dynamically compressed parallel to the X_1 or X_2 -axis (these are equivalent) while being imaged on a plane normal to X_3 . The obtained stress–time history and corresponding video frames are shown in Figure 8. The carbonaceous inclusions now appear to have larger cross-sections and are more randomly oriented with respect to the compression axis. A peak compressive stress of 3.4 GPa was recorded at time t_7 . Similar to the observations from Figure 7, microcracks were found to initiate around the carbonaceous inclusions prior to peak stress. An example of such microcrack initiation process is shown in the inset at time t_1 . Comparing this inset to an earlier time of t^* shows a pair of microcracks that are formed in close proximity of the inclusion. Ejecta cloud formation was again a prevalent failure process observed as early as time t_2 from the marked region. The ejecta can be better visualized in the video recording of this experiment which can be accessed by following the link provided for Video 2 in Section S1. In this orientation, formation of axial microcracks and localized fragmentation appear to occur nearly simultaneously beginning at time t_3 . This implies that additional cracks were likely present underneath the surface which coalesced with the newly formed axial cracks and resulted in fragmentation. The interaction of microcracks with inclusions is observed at times t_4 and t_5 when an axial crack reaches an inclusion and passes through it. After the interaction, the initial crack is still visible along the same trajectory and a secondary crack which is much

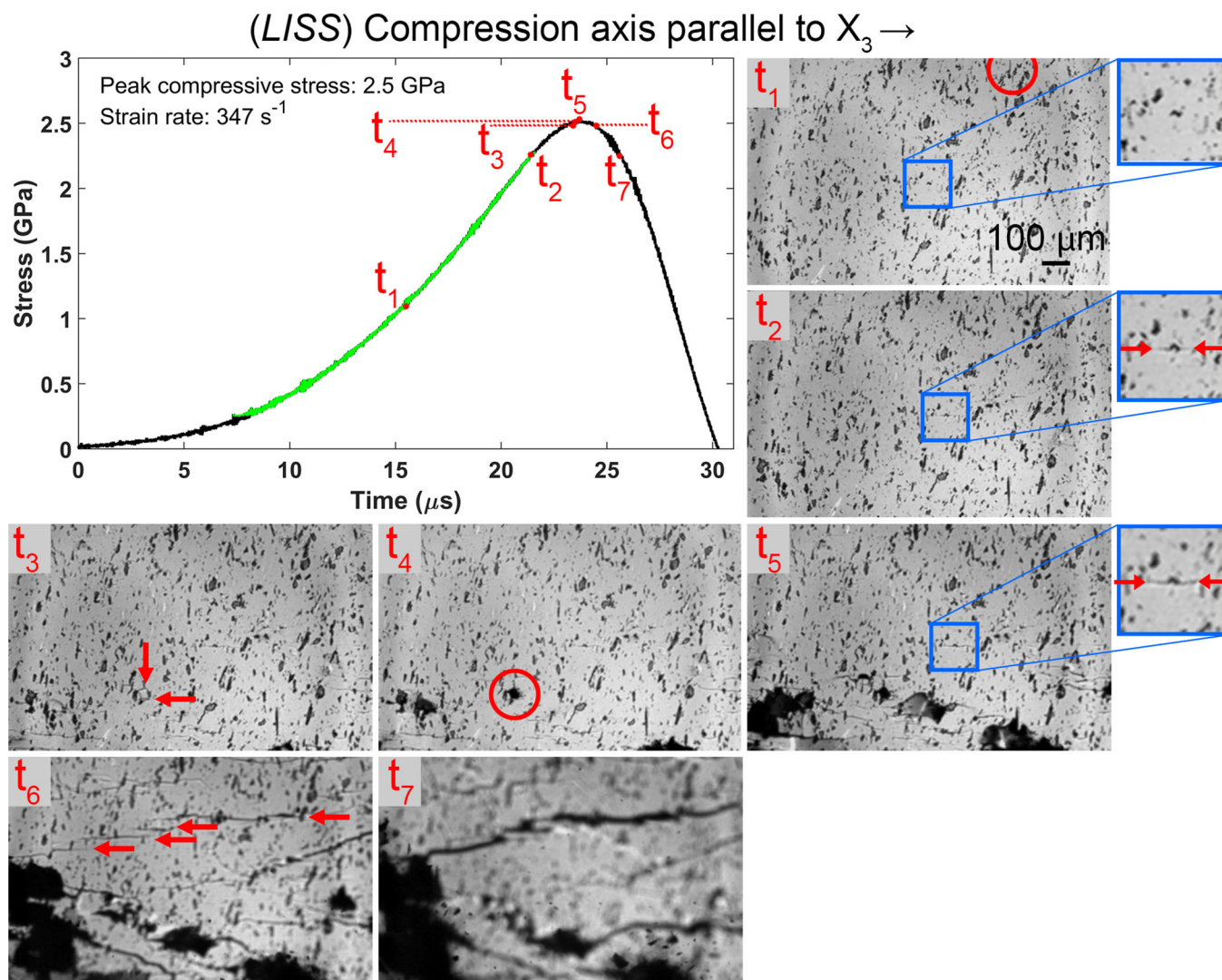


FIGURE 7 Stress–time history and time-resolved snapshots for the L-inclusion S-spacing *targeted* microstructure. The compression axis was parallel to the hot-pressing direction. High-speed imaging was performed through a 10 $\times$ microscope objective lens. Insets show a carbonaceous inclusion that initiates a set of microcracks at time t_2 and arrows denote the position of cracks.

wider also forms, leading to fragmentation. Formation of axial cracks and fragmentation continue to occur and eventually at time t_6 inclined cracks that are $\sim 45^\circ$ with respect to the compression axis appear; these rapidly coalesce with the existing axial cracks and create larger bulky fragments. The fragments start to drift outside of the depth of field, but their overall dimensions appear to be preserved until peak stress. The stress and strain rates for this experiment (211 MPa/ μm and 507 s^{-1} , respectively) are slightly higher than those obtained from the experiment shown in Figure 7, where the compression axis was parallel to the X_3 .

Unlike previous reports from Kolsky bar experiments on PAD B_4C where the majority of failure processes were observed near or after peak stress,^{10,14} incorporating the microscope objective with the high-speed camera in this study allowed for visualization of early-time failure pro-

cesses during dynamic deformation. The formation and ejection of small fragments was one of the first processes observed in the both studied orientations. The ejected material was seen to form from the regions that contain carbonaceous inclusions. SEM micrographs of the inclusions shown in Figure S8 suggest that the interior graphite platelets are still loosely bonded. It is thus plausible that the sliding and dissociation of the loosely-packed graphite platelets upon the arrival of the stress pulse triggers the material ejection. We note that ejecta formation may have been facilitated by the mechanical polishing of the imaged surface which had exposed the interior of the inclusions. Localized fragmentation was another failure process that was observed at later times and resulted in formation of microscale fragments. Coalescence of axial and transverse microcracks from neighboring inclusions were found to

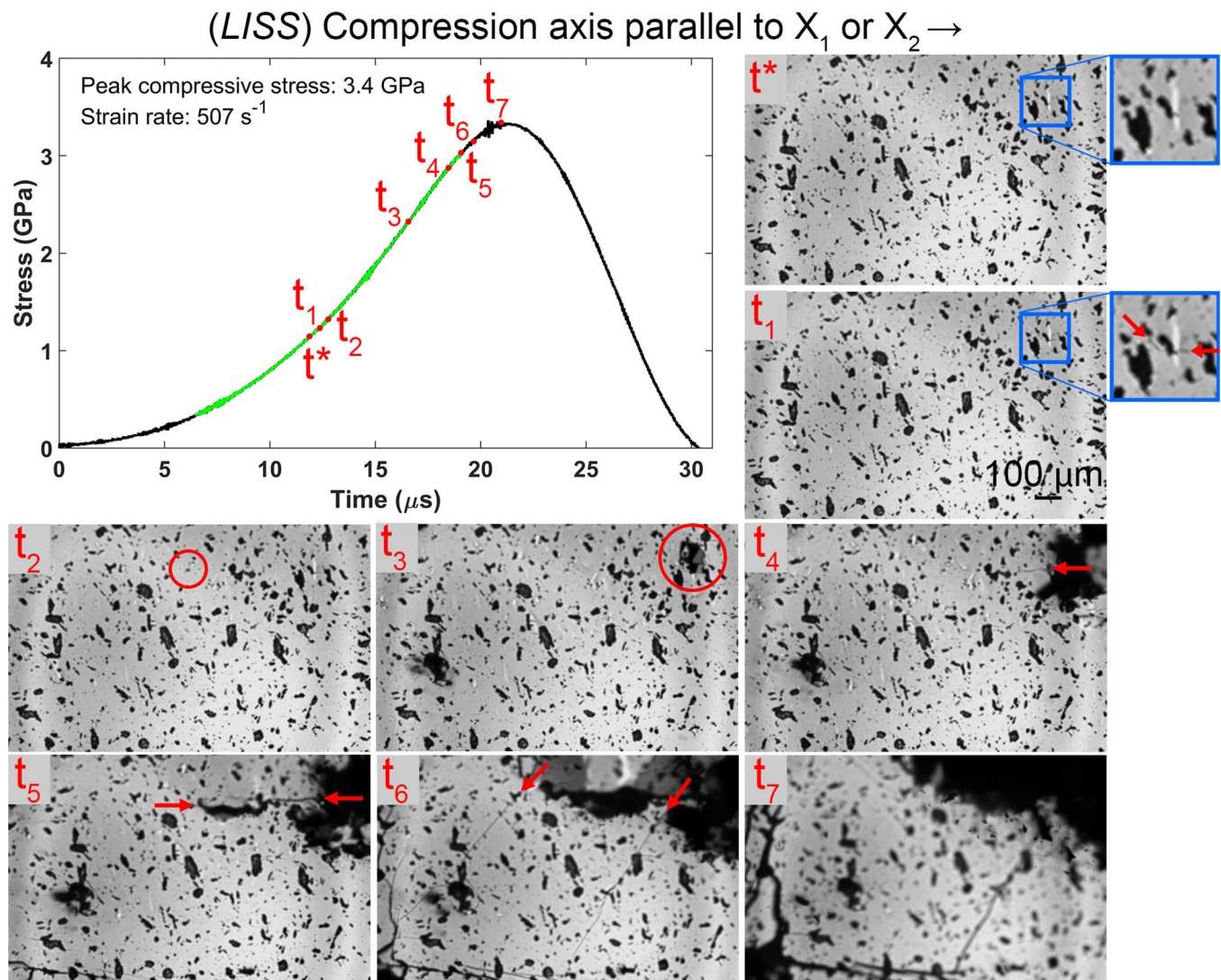


FIGURE 8 Stress–time history and time-resolved snapshots for the L-inclusion S-spacing *targeted* microstructure. The compression axis was perpendicular to the hot-pressing direction. High-speed imaging was performed through a 10× microscope objective lens. Insets show a carbonaceous inclusion that initiates a set of microcracks at time t_1 and arrows denote the position of cracks.

be mechanisms through with localized fragmentation was accommodated.

To correlate the localized failure processes observed in Figures 7 and 8 with the structural failure of the prismatic specimens, we turn our attention to the experiment shown in Figure 9, where lower-resolution imaging is used so that the entire specimen is visible in the field of view. In this experiment, the compression axis was parallel to X_3 and imaging was performed on a plane normal to X_1 or X_2 which had a 1200 grit final finish. A peak compressive stress of 2.6 GPa was achieved at time t_5 and the stress and strain rates were 162 MPa/μm and 383 s⁻¹, respectively. The earliest failure processes were observed at time t_2 in the form of edge failure (marked by arrows) and localized fragmentation (marked by circles). Both types of failure processes were found to be simultaneously active at later

times (t_3) with the accumulated damage eventually resulting in structural failure as seen at times t_4 and t_5 . Edge failure has been commonly reported during Kolsky bar experiments of ceramics^{10,14} and is believed to be resultant from lateral tensile stresses that develop at the specimen/platen interfaces. Various experimental issues could contribute to edge failure, including imperfect contact or unintended frictional forces at the specimen interfaces and eccentric loading.^{34,37} The axial cracks that develop as a result of these issues are less likely to be correlated with the microcracks that initiated from the carbonaceous inclusions. In contrast, localized fragmentation is believed to be the consequence of the coalescence of microcracks that are initiated from the carbonaceous inclusions. This argument is supported by the experimental observations of Figures 7 and 8 (see the corresponding frames at times t_3 – t_4 and t_3 ,

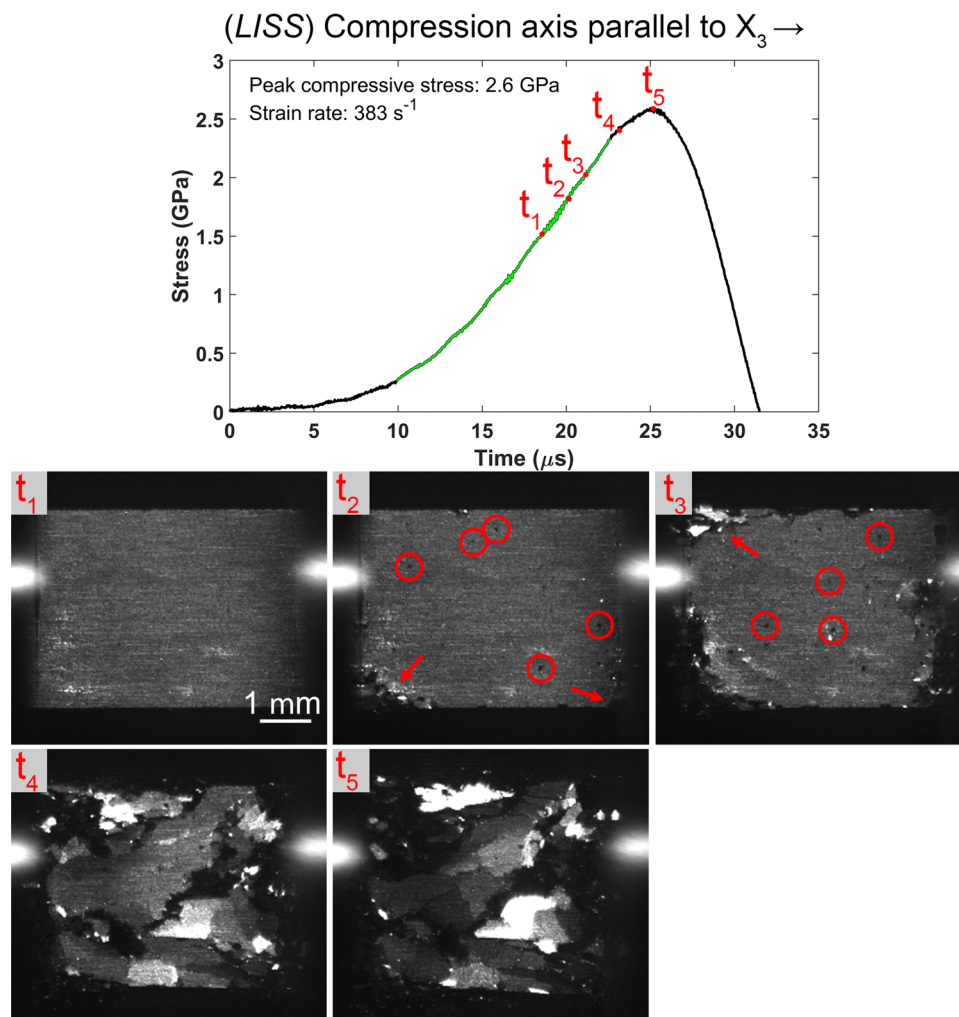


FIGURE 9 Stress–time history and time-resolved snapshots for the L-inclusion S-spacing *targeted* microstructure. The compression axis was parallel to the hot-pressing direction.

respectively). Even though it is not possible to determine quantitatively the degree to which edge failure and microcracking contribute to structural failure, these observations demonstrate that the carbonaceous inclusions facilitate the dynamic failure of boron carbide. We also note that it is likely for the two failure processes to interact as the axial macrocracks start to propagate across the specimen.

Kolsky bar experiments were also performed on the NFC material to study the failure processes in the absence of carbonaceous inclusions. Figure 10 shows the stress–time history and corresponding video frames obtained from the NFC material when the compression axis was parallel to the X_3 -axis and imaging was performed on a plane normal to X_1 or X_2 . Note that even though the magnification is the same as Figures 7 and 8, no inclusions are visible in this case. A peak compressive stress of 4.9 GPa was obtained at time t_1 . Unlike the LISS material, no failure processes were observed prior to peak stress. The first axial crack enters the field of view at time t_2 and grows in an unstable man-

ner with minimum deflection. At time t_3 , more axial and inclined cracks are observed, some of which appear to be initiated from the inside of the field of view. Fragmentation is observed at times t_4 and t_5 . Eventually, transverse cracks that are nearly perpendicular to the compression axis are observed at time t_6 . Coalescence of the transverse and axial cracks results in formation of larger fragments that drift outside of the depth of field while maintaining their overall dimensions up to time t_7 . The stress and strain rates for this experiment were estimated to be $250 \text{ MPa}/\mu\text{m}$ and 468 s^{-1} , respectively. The modulus of the NFC material was assumed to be equal to the value obtained for the hard constituent phase by clustering the indentation data (see Table 5). The effect of orientation on the NFC material was studied by dynamically compressing a specimen parallel to the X_1 - or X_2 -axis while conducting imaging on a plane normal to X_3 . The results are shown in Figure 11. A peak compressive stress of 4.6 GPa was observed at time t_1 with no failure processes visible on the surface up to or at that

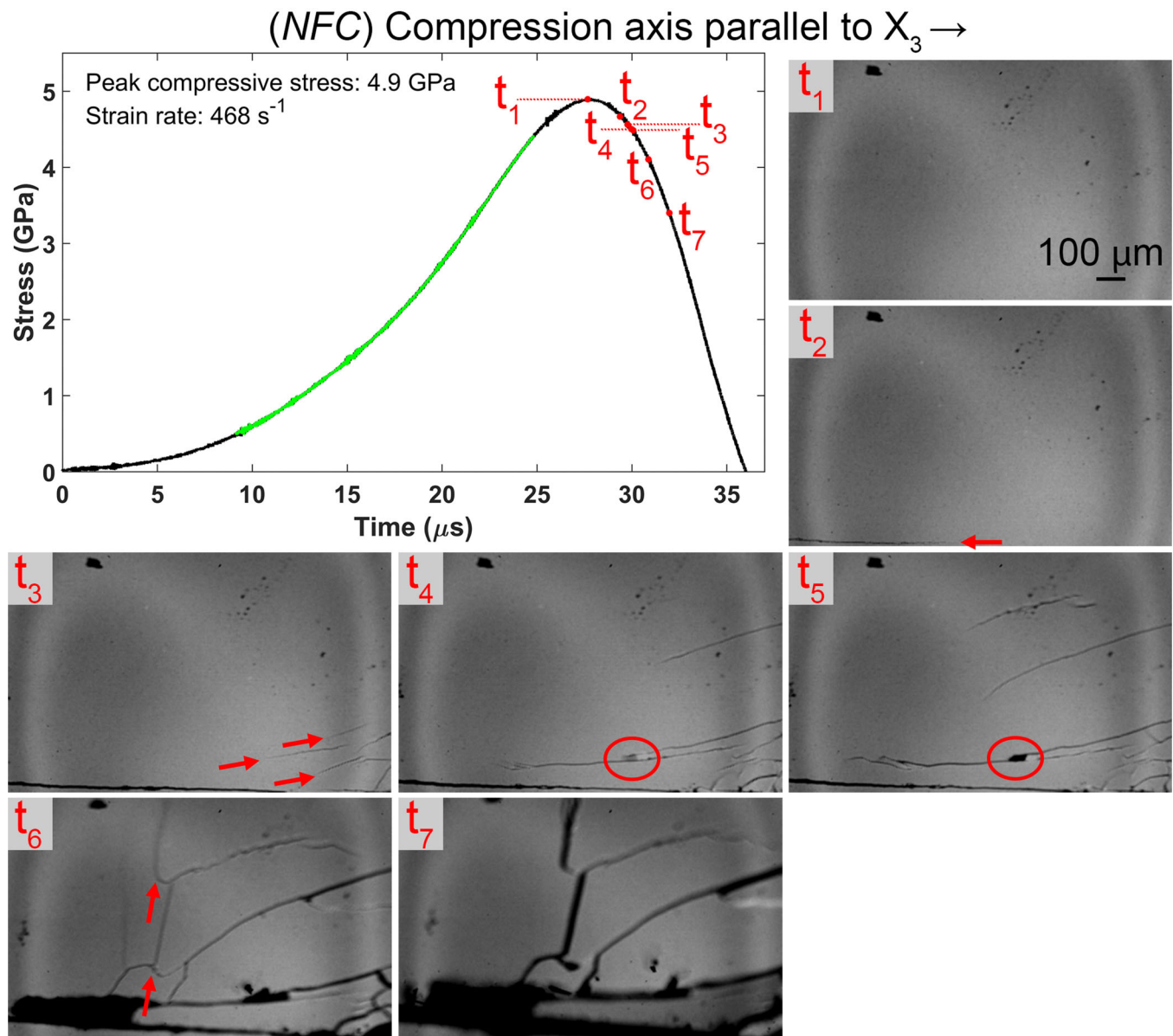


FIGURE 10 Stress–time history and time-resolved snapshots for the NFC material. The compression axis was parallel to the hot-pressing direction. High-speed imaging was performed through a 10 $\times$ microscope objective lens. NFC, no free carbon.

time. The first axial crack enters the field of view at time t_2 , and propagates in an unstable manner creating smaller fragments along its path as seen at time t_3 . Additional axial cracks are observed at time t_4 which are then linked to the previously formed axial cracks through formation of transverse cracks at time t_5 . As fragmentation in the lower region of the field of view continues, more cracks are formed in the upper region, the majority of which are either inclined or perpendicular to the compression axis as seen at time t_6 . Some of these cracks change direction during propagation and orient themselves approximately parallel to the loading direction at time t_7 , eventually splitting into two transverse cracks at time t_8 , and resulting in fragmentation in the upper region of the field of view. The

stress and strain rates for this experiment ($234 \text{ MPa}/\mu\text{m}$ and 436 s^{-1} , respectively) are comparable to the values obtained in the previous experiment (shown in Figure 10), where the compression axis was parallel to the X_3 .

In both orientations of the NFC material, the first observed axial crack was initiated from outside of the field of view. To better understand crack initiation in the NFC material, we consider the experiment shown in Figure 12 where imaging was performed without the microscope objective. A peak compressive stress of 5.1 GPa, stress rate of $292 \text{ MPa}/\mu\text{m}$, and strain rate of 545 s^{-1} were obtained in this experiment. Failure around the top edge of the specimen is observed at the time of peak stress (t_1). Beyond peak stress at times t_2 – t_5 , edge failure is observed and the

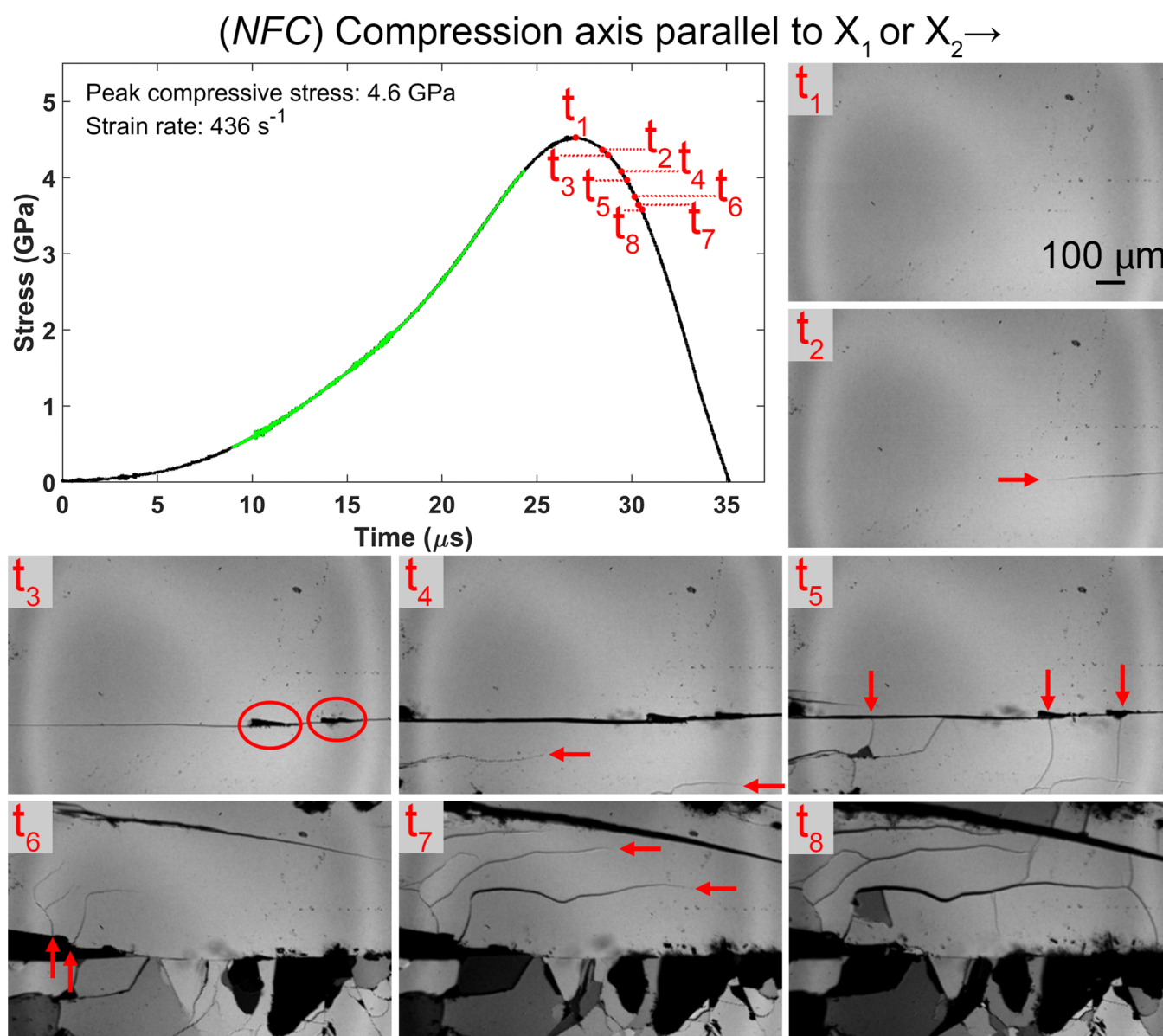


FIGURE 11 Stress–time history and time-resolved snapshots for the NFC material. The compression axis was perpendicular to the hot-pressing direction. High-speed imaging was performed through a 10 $\times$ microscope objective lens. NFC, no free carbon.

resultant span-wise axial macrocracks lead to the formation of several parallel columnar fragments. This is in contrast to the crack propagation behavior observed in Figure 9, where the span-wise propagation of the axial macrocracks was hindered by localized fragmentation. Eventually at time t_6 , transverse macrocracks are formed due to buckling of the columnar fragments.³⁸ We also note that localized fragmentation was also observed in

one experiment on the NFC material (see Figure S9) and resulted in formation of millimeter-sized fragments. A plausible mechanism for localized fragmentation in the NFC material is crack initiation from the sub-micron inclusion clusters that were sporadically observed in the microstructure of the NFC material (see Section 3.1). In light of these observations, it is likely that the first axial cracks observed in Figures 10 and 11 were also

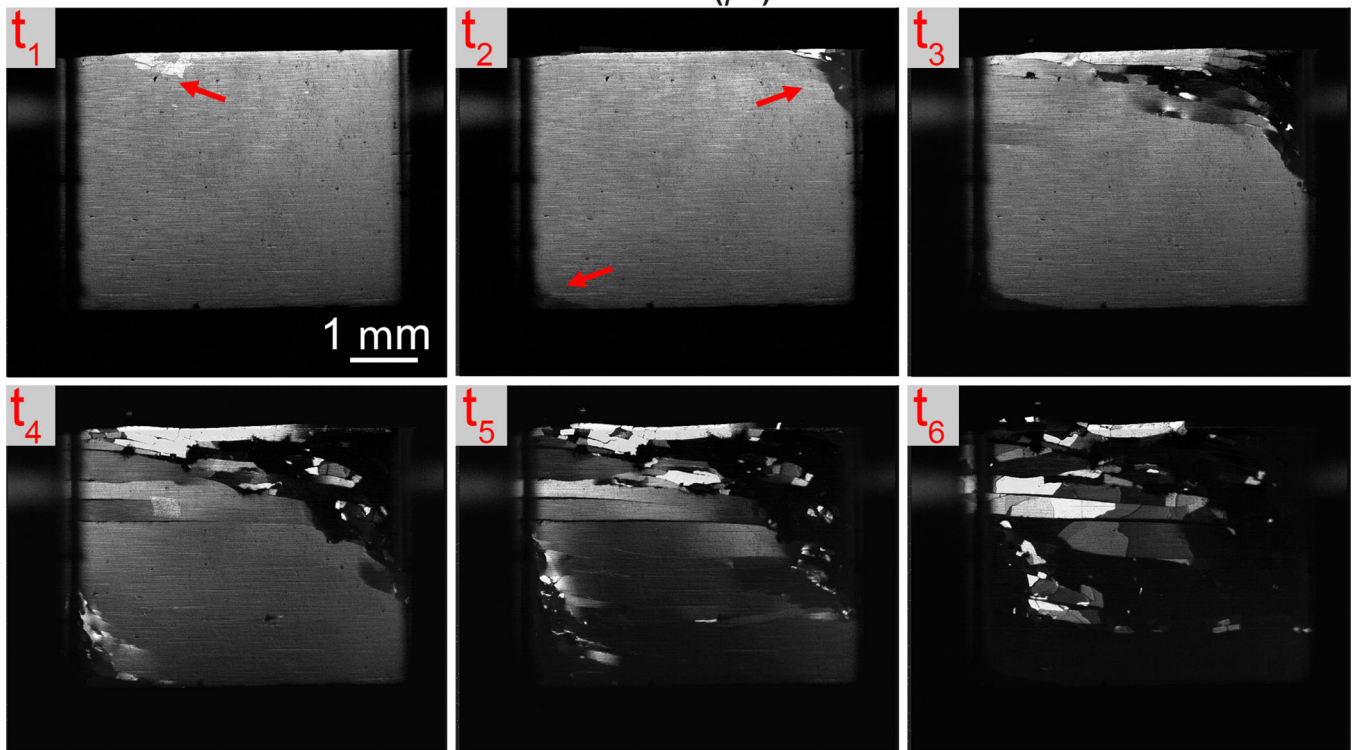
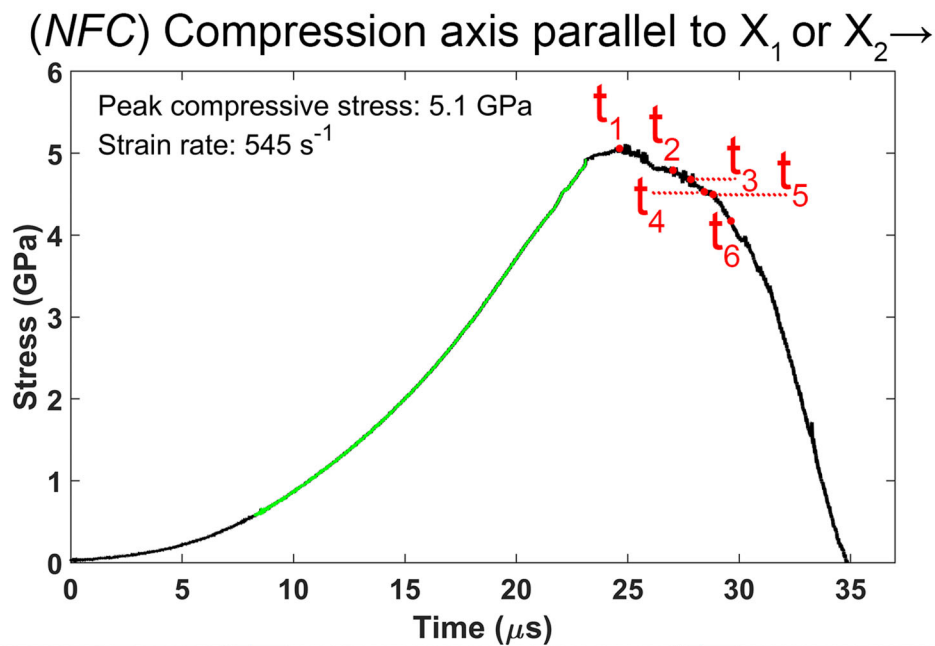


FIGURE 12 Stress–time history and time-resolved snapshots for the NFC material. The compression axis was perpendicular to the hot-pressing direction. NFC, no free carbon.

initiated by edge failure and thus are not correlated with the microstructure.

4.3.2 | Variations of dynamic peak stress

Figure 13A,B show the dependence of the dynamic peak compressive stress on the MLE values of size and spacing of

the carbonaceous inclusions for the in-plane and through-plane orientations, respectively. The overall range of stress values obtained from testing the SILS, SISS, LILS, and LISS *targeted* microstructures was 1.8–5.4 GPa. We note that the lower peak stress values may be an artifact of the prismatic specimen geometry used in this study which could generate stress concentrations and result in premature failure and lower peak stress values by as much as 30%.^{34,37,39}

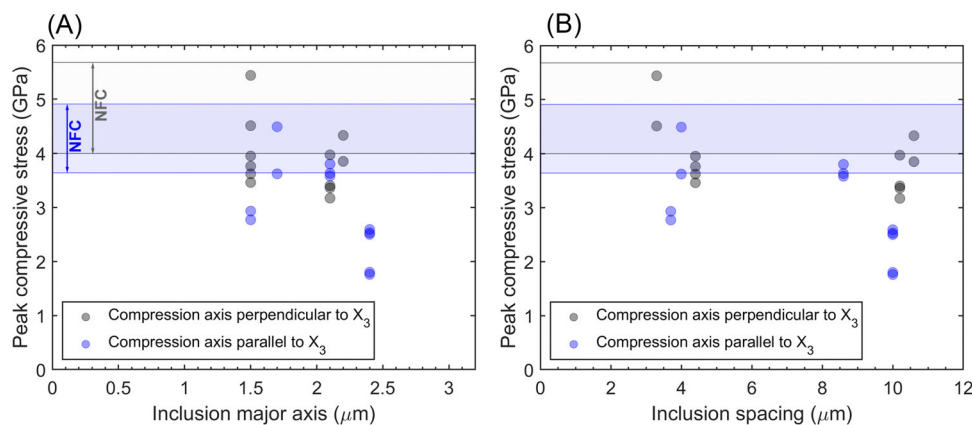


FIGURE 13 Variations of the peak compressive stress from Kolsky bar experiments with respect to the (A) size and (B) spacing of the carbonaceous inclusions. The shaded bands represent the range of values obtained for the NFC material. NFC, no free carbon.

Furthermore, a significant overlap was observed between the resultant microstructures (as discussed in Section 4.1) and the total number of peak stress data points was also small (12 for each orientation). In light of these uncertainties, we do not attempt to deconvolve the explicit role of inclusion size or spacing on peak stress. Instead, we turn our attention to the differences between the materials that contain carbonaceous inclusions and the NFC material for which the obtained peak stress values varied from 3.6 to 5.7 GPa. The range of peak stress values for the in-plane and through-plane orientations of the NFC material are represented by the shaded bands in Figure 13A,B. Removal of the carbonaceous inclusions is seen to result in an overall increase of the peak stress. Delayed crack initiation in the absence of intrinsic flaws (as evidenced by the observations of the dynamic failure processes presented in Section 4.3.1) is a plausible explanation for the higher peak stress of the NFC material.

Figure 13 also shows that the peak stress is orientation-dependent with the stress being lower along the X_3 -axis (i.e., hot-pressing direction). This was also observed in PAD B_4C with a peak stress of 4.5 and 3.8 GPa reported perpendicular and parallel to the hot-pressing direction, respectively.¹³ For PAD B_4C , this was attributed to the preferential orientations of disk-shaped carbonaceous inclusions which made the inclusion orientations less favorable for crack initiation in the in-plane orientation.^{13,14} However, comparing the bivariate KDE plots of Figure 5 does not support this argument for the materials studied here. Furthermore, the NFC material also exhibits a similar anisotropy in peak stress, which implies that characteristics other than the inclusion distribution (e.g., grain anisotropy or texture) can lead to the observed anisotropy. The anisotropy in peak stress of the NFC material was further studied by evaluating the crack tip velocities (v_C) in the through-plane and in-plane orientations. The first

axial cracks observed in the experiments of Figures 10 and 11 were used for these measurements. To facilitate the comparison, the computed crack tip velocities were normalized by the corresponding Rayleigh wave speed (v_R) approximated by⁴⁰

$$v_R = \sqrt{\frac{G}{\rho}} \left(\frac{0.862 + 1.14\nu_s}{1 + \nu_s} \right) \quad (12)$$

with G and ρ representing the shear modulus (assumed to be 197 GPa²²) and density (2.5 g/cm³²²) of boron carbide. As seen in Figure 14, the normalized crack tip velocity was found to be higher for loading along the hot-pressing direction, which is consistent with the lower peak stress obtained in the through-plane orientation. Note that the crack speeds decrease with time during our observations. Although the observed crack speeds are very high, these speeds are consistent with our expectations from dynamic fracture theory. We note that even though the computed crack tip velocities change slightly based on the threshold values used for image segmentation, the relative differences between the two orientations remain unchanged.

5 | DISCUSSION

In this study, some input variables in the hot-pressing process were tuned to engineer four microstructures with two output parameters: size and spacing of the resultant carbonaceous inclusions. First, different sizes of the initial graphite powders were selected to maximize the likelihood of obtaining either small or large inclusions. The as-received powders were also sieved to further narrow down the range of particle sizes. Second, the concentration of each type of graphite powder was varied to obtain either

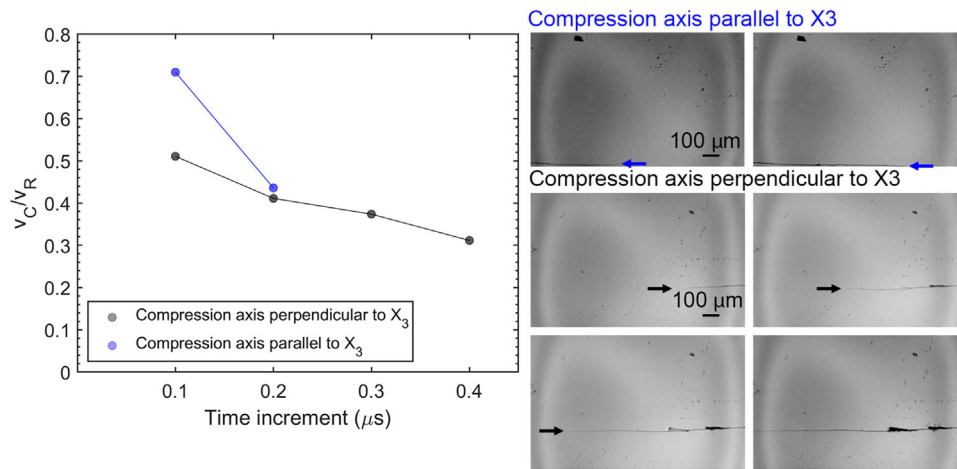


FIGURE 14 Crack tip velocity normalized to the Rayleigh wave speed (v_R) for the NFC material. The corresponding frames used for evaluating velocities are shown on the right. NFC, no free carbon.

small or large spacings between the inclusions. Characterization of the resultant inclusion sizes showed a higher upper range and mean for the materials prepared with the larger graphite powder. However, the lower range of inclusion sizes was comparable for all the materials. Thus, a significant overlap exists between the inclusion sizes of the S- versus L-inclusion *targeted* microstructures. In terms of spacings between the inclusions, materials that were prepared with different concentrations of the same graphite powder had comparable mean spacings and inclusion number densities. Differences were observed between the probability distribution functions of the inclusion spacings but the trend was not consistent among different materials. Characterization of inclusion orientations also showed a strong correlation between the size and orientation of the inclusions, suggesting that it may not be appropriate to model some inclusion attributes as independent variables. Overall, the differences between the microstructures were insufficient to make strong statements about the quantitative effects of the intrinsic flaws. This highlights the role of uncertainties involved in microstructural engineering of advanced ceramics. Identifying the sources of uncertainty and their management are essential for reaching improved material property spaces through the materials-by-design approach.

5.1 | Influence of intrinsic flaws on dynamic compressive failure

Similar to previous reports on commercially available PAD B₄C,¹⁴ carbonaceous inclusions were found to be the first constituent phase to undergo deformation and to act as nucleation sites for wing cracks. The formation of fine clouds of ejecta also suggests that it may be more appro-

priate to model the carbonaceous inclusions as disks filled with granular media rather than simply as soft inclusions.

Real-time observations of the failure processes presented in Figure 7 provide a clear view of wing crack initiation. Wing cracks are formed prior to peak stress and grow in population as the stress rises. Many of the nucleated wing cracks travel only a short distance before their propagation is arrested. Noting the cross-sections of the inclusions that served as crack nucleation sites suggests that they are among the smaller ones observable in the field of view. This is in agreement with the findings of Figure 5 which showed that the likelihood of finding favorably oriented inclusions decreased as the size of the inclusion increased, that is, the larger inclusions were less likely to be appropriately oriented for crack initiation. Given the observed overlap in the size of appropriately oriented inclusions in the four *targeted* microstructures (SILS, SISS, LILS, and LISS), the crack initiation process in these materials is expected to be similar. Note that Hogan et al. observed that in PAD B₄C larger carbonaceous inclusions were the most favorable nucleation sites for cracks.¹⁰ This contrasting behavior could be because of differences between the inclusion size and orientation distributions in the materials examined by the two studies. We note also that the higher spatial and temporal resolutions in our study (due to SEM-based rather than optical characterization of the microstructures, and the new ultra-high-speed microscopy approach) provide real-time observations of crack initiation instead of relying on post-mortem examinations. As the stress approaches the peak value, we observe that wing cracks that are in close proximity coalesce to form unstable axial cracks that lead to structural failure. At later times, transverse cracks are also formed as a result of the buckling of these columnar fragments.

In the absence of carbonaceous inclusions (i.e., the NFC material), wing cracks are no longer an active failure process and majority of the axial cracks are initiated at the specimen/bar interfaces and result in axial splitting of the specimen into several parallel columnar fragments which then buckle and form transverse cracks. Thus, it appears that in the absence of intrinsic flaws, dynamic failure is mainly mediated by crack initiation from structural imperfections like sharp corners and edges that create stress concentrations.

6 | CONCLUSIONS

In this study, boron carbide ceramics with controlled distributions of carbonaceous inclusions were investigated to understand the effects of such heterogeneities on the dynamic failure processes of advanced engineering ceramics. Detailed statistical characterization of the microstructures of these materials demonstrated that the size and spacing distributions had significant overlap. Ultra-high-speed microscopic observations of the microscale failure processes of these materials during dynamic loading, performed in situ for the first time, demonstrated that the initiation of wing cracks is controlled by the populations of carbonaceous inclusions. For the microstructures and geometries investigated in this study, peak compressive stress was increased after the removal of these processing-controlled heterogeneities. In the absence of such intrinsic flaws, geometric stress concentrations and structural design of the specimens were found to be the major contributors to the dynamic failure of the ceramic. Dynamic peak compressive stress was found to be anisotropic regardless of the presence of the carbonaceous inclusions.

The experimental data obtained in this study are expected to be valuable for calibrating and validating models of brittle fracture in advanced ceramics. The findings provide valuable insights into processing–microstructure–property correlations and the involved uncertainties and demonstrate that removal of intrinsic flaws could be a potential pathway toward improving the dynamic mechanical properties of boron carbide ceramics.

ACKNOWLEDGMENTS

Research was sponsored by The Army Research Laboratory and was accomplished under Cooperative Agreement Number W911NF-12-2-0022. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the Army Research Laboratory or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for Government purposes notwithstanding any copyright notation

herein. Some of the writing was done under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under Contract DEAC52-07NA27344.

The authors acknowledge the help of Dheeraj Dhanvee Kairamkonda and Kevin Hu with segmentation of the SEM results, as is the help of Levi Johnson with the Kolsky bar experiments, and Kyle Fisher with some of the specimen preparation. The contributions of Lori Graham-Brady and Amartya Bhattacharjee towards the conception and initiation of this study as well as the guidance of James W. McCauley on interpretation of the results are also greatly acknowledged.

ORCID

Arezo Zare  <https://orcid.org/0000-0002-2481-3626>

Kent Christian  <https://orcid.org/0000-0003-0783-4799>

REFERENCES

- Basu B, Balani K. Advanced structural ceramics. New Jersey: John Wiley and Sons; 2011.
- Padture NP. Advanced structural ceramics in aerospace propulsion. *Nat Mater*. 2016;15:804–9.
- Bavdekar S, Subhash G. Failure mechanisms of ceramics under quasi-static and dynamic loads: overview. In: Voyiadis GZ, editor. *Handbook of damage mechanics*. Switzerland: Springer; 2022. p. 579–607.
- Subhash G, Awasthi A, Ghosh D. Dynamic response of advanced ceramics. New Jersey: John Wiley and Sons; 2021.
- Ramesh KT, Hogan JD, Kimberley J, Stickle A. A review of mechanisms and models for dynamic failure, strength, and fragmentation. *Planet Space Sci*. 2015;107:10–23.
- Chen M, McCauley JW, Hemker KJ. Shock-induced localized amorphization in boron carbide. *Science*. 2003;299:1563–6.
- Reddy KM, Guo JJ, Shinoda Y, Fujita T, Hirata A, Singh JP, et al. Enhanced mechanical properties of nanocrystalline boron carbide by nanoporosity and interface phases. *Nat Commun*. 2012;3:1–7.
- Zhao S, Kad B, Remington BA, LaSalvia JC, Wehrenberg CE, Behler KD, et al. Directional amorphization of boron carbide subjected to laser shock compression. *Proc Natl Acad Sci*. 2016;113:12088–93.
- Jeong B, Lahkar S, An Q, Reddy KM. Mechanical properties and deformation behavior of superhard lightweight nanocrystalline ceramics. *Nanomaterials*. 2022;12:3228.
- Hogan JD, Farbaniec L, Shaeffer M, Ramesh KT. The effects of microstructure and confinement on the compressive fragmentation of an advanced ceramic. *J Am Ceram Soc*. 2015;98:902–12.
- Farbaniec L, Hogan JD, Xie KY, Shaeffer M, Hemker KJ, Ramesh KT. Damage evolution of hot-pressed boron carbide under confined dynamic compression. *Int J Impact Eng*. 2017;99:75–84.
- Farbaniec L, Hogan JD, Ramesh KT. Micromechanisms associated with the dynamic compressive failure of hot-pressed boron carbide. *Scr Mater*. 2015;106:52–6.
- Farbaniec L, Hogan J, McCauley J, Ramesh KT. Anisotropy of mechanical properties in a hot-pressed boron carbide. *Int J Appl Ceram Technol*. 2016;13:1008–16.

14. Hogan JD, Farbaniec L, Sano T, Shaeffer M, Ramesh KT. The effects of defects on the uniaxial compressive strength and failure of an advanced ceramic. *Acta Mater.* 2016;102:263–72.
15. Sun X, Chauhan A, Mallick DD, Tonge AL, McCauley JW, Hemker KJ, et al. Granular flow of an advanced ceramic under ultra-high strain rates and high pressures. *J Mech Phys Solids.* 2020;143:104031.
16. Ghosh D, Subhash G, Zheng JQ, Halls V. Influence of stress state and strain rate on structural amorphization in boron carbide. *J Appl Phys.* 2012;111:063523.
17. An Q, Goddard III WA, Cheng T. Atomistic explanation of shear-induced amorphous band formation in boron carbide. *Phys Rev Lett.* 2014;113:095501.
18. An Q, Goddard III WA. Atomistic origin of brittle failure of boron carbide from large-scale reactive dynamics simulations: suggestions toward improved ductility. *Phys Rev Lett.* 2015;115:105501.
19. DeVries M, Subhash G, Awasthi A. Shocked ceramics melt: an atomistic analysis of thermodynamic behavior of boron carbide. *Phys Rev B.* 2020;101:144107.
20. Reddy KM, Guo D, Song S, Cheng C, Han J, Wang X, et al. Dislocation-mediated shear amorphization in boron carbide. *Sci Adv.* 2021;7:eabc6714.
21. Blumenthal WR, Gray GT, Claytor TN. Response of aluminium-infiltrated boron carbide cermets to shock wave loading. *J Mater Sci.* 1994;29:4567–76.
22. Ramesh KT, Graham-Brady L, Goddard WA, Hurley RC, Robbins M, Tonge AL, et al. Models for the behavior of boron carbide in extreme dynamic environments. *J Am Ceram Soc.* 2021;105:3043–61.
23. Kimberley J, Ramesh KT, Daphalapurkar NP. A scaling law for the dynamic strength of brittle solids. *Acta Mater.* 2013;61:3509–21.
24. Daphalapurkar NP, Ramesh KT, Graham-Brady L, Molinari JF. Predicting variability in the dynamic failure strength of brittle materials considering pre-existing flaws. *J Mech Phys Solids.* 2011;59:297–319.
25. Moorehead CA, Sietins JM, Swab JJ. Meso-scale microstructural agglomerate quantification in boron carbide using X-ray microcomputed tomography. *Mater Charact.* 2018;141:177–85.
26. Kuwelkar KA. Chemical and structural characterization of boron carbide powders and ceramics. Ph.D. thesis, Rutgers University; 2016.
27. Zare A, He MR, Straker M, Chandrashekhare MV, Spencer M, Hemker KJ, et al. Mechanical characterization of boron carbide single crystals. *J Am Ceram Soc.* 2022;105:3030–42.
28. Domnich V, Reynaud S, Haber RA, Chhowalla M. Boron carbide: structure, properties, and stability under stress. *J Am Ceram Soc.* 2011;94:3605–28.
29. Malhotra P, Niu S, Srivastava V, Guduru PR. A technique for high-speed microscopic imaging of dynamic failure events and its application to shear band initiation in polycarbonate. *J Appl Mech.* 2022;89:041001.
30. Feathers JR, Spoth KA, Fromme JC. Surpassing the physical Nyquist limit to produce super-resolution cryo-EM reconstructions. *bioRxiv.* 2020;675397.
31. Nemat-Nasser S, Horii H. Compression-induced nonplanar crack extension with application to splitting, exfoliation, and rockburst. *J Geophys Res Solid Earth.* 1982;87:6805–21.
32. Zeng Q, Tonge AL, Ramesh KT. A multi-mechanism constitutive model for the dynamic failure of quasi-brittle materials. Part II: Integrative model. *J Mech Phys Solids.* 2019;131:20–42.
33. Vargas-Gonzalez L, Speyer RF, Campbell J. Flexural strength, fracture toughness, and hardness of silicon carbide and boron carbide armor ceramics. *Int J Appl Ceram Technol.* 2010;7:643–51.
34. Swab JJ, Meredith CS, Casem DT, Gamble WR. Static and dynamic compression strength of hot-pressed boron carbide using a dumbbell-shaped specimen. *J Mater Sci.* 2017;52:10073–84.
35. Pradhan SK, Nayak BB, Sahay SS, Mishra BK. Mechanical properties of graphite flakes and spherulites measured by nanoindentation. *Carbon.* 2009;47:2290–2.
36. Richter A, Ries R, Smith R, Henkel M, Wolf B. Nanoindentation of diamond, graphite and fullerene films. *Diam Relat Mater.* 2000;9:170–84.
37. Swab JJ, Chen W, Hogan J, Liao Ha, Lo C, Mates S, et al. Dynamic compression strength of ceramics: what was learned from an interlaboratory round robin exercise? *J Dyn Behav Mater.* 2021;7:34–47.
38. Ashby MFA, Hallam SD. The failure of brittle solids containing small cracks under compressive stress states. *Acta Metall.* 1986;34:497–510.
39. Blumenthal WR. High strain rate compression testing of ceramics and ceramic composites. *Ceram Eng Sci Proc.* 2005;26:89–96.
40. Freund LB. Dynamic fracture mechanics. Cambridge, UK: Cambridge University Press; 1998.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Zare A, Malhotra P, Yang Q, Christian K, Du J, Haber RA, et al. The effects of carbonaceous inclusions and their distributions on dynamic failure processes in boron carbide ceramics. *J Am Ceram Soc.* 2023;1–21. <https://doi.org/10.1111/jace.19101>